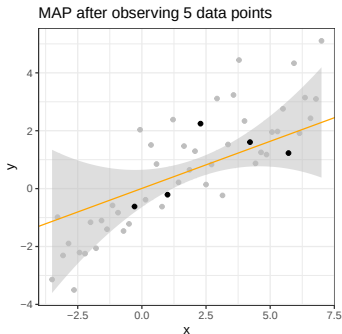
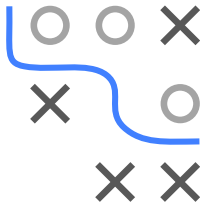


Introduction to Machine Learning

Gaussian Processes

Bayesian Linear Model

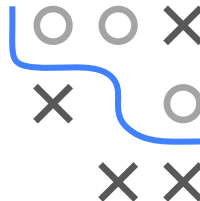
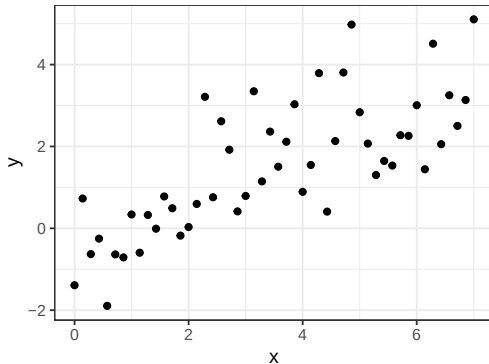


Learning goals

- Know the Bayesian linear model
- The Bayesian LM returns a (posterior) distribution instead of a point estimate
- Know how to derive the posterior distribution for a Bayesian LM

DATA SITUATION

- $\mathcal{D} = ((\mathbf{x}^{(1)}, y^{(1)}), \dots, (\mathbf{x}^{(n)}, y^{(n)}))$: i.i.d. training set from some unknown distribution



- $\mathbf{X} \in \mathbb{R}^{n \times p}$: design matrix, where i -th row contains vector $\mathbf{x}^{(i)}$
- $\mathbf{y} = (y^{(1)}, \dots, y^{(n)})^\top$

BAYESIAN LINEAR MODEL REVISITED

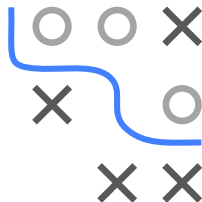
- Standard linear regression model for i -th observation, with $\theta \in \mathbb{R}^p$ fixed but unknown

$$y^{(i)} = f(\mathbf{x}^{(i)}) + \epsilon^{(i)} = \theta^T \mathbf{x}^{(i)} + \epsilon^{(i)} \quad \forall i$$

- Assumption: function outputs $f(\mathbf{x}^{(i)})$ differ from observed values $y^{(i)}$ by additive, i.i.d. Gaussian noise (ind of $\mathbf{x}, \theta$)

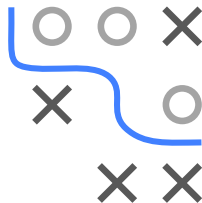
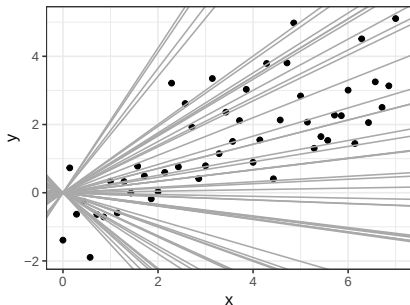
$$\epsilon^{(i)} \sim \mathcal{N}(0, \sigma^2) \quad \forall i$$

- Bayesian perspective: θ also RV with associated (prior) distribution, e.g., $\theta \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_p)$



GP PERSPECTIVE

- Weight-space view: prior over θ , function-space view: prior over linear functions
- Example: random lines with intercept 0, slope $\theta \sim \mathcal{N}(0, 1)$



- Random lines = draws from GP with linear kernel
- Collection of RVs $\{f(\mathbf{x}) = \theta\mathbf{x} : \mathbf{x} \in \mathbb{R}\}$
- $f(\mathbf{X})$ is mv Gaussian for any finite input with design matrix $\mathbf{X}$

FROM PRIOR TO POSTERIOR

- Bayes' rule: update prior to posterior belief after observing data

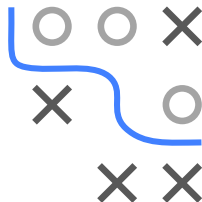
$$p(\theta|\mathbf{X}, \mathbf{y}) = \frac{\text{likelihood} \cdot \text{prior}}{\text{marginal likelihood}} = \frac{p(\mathbf{y}|\mathbf{X}, \theta) \cdot q(\theta)}{p(\mathbf{y}|\mathbf{X})}$$

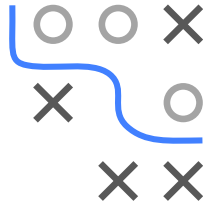
- Gaussian family is “self-conjugate”: Gaussian prior & Gaussian likelihood $\Rightarrow$ Gaussian posterior

$$\theta \mid \mathbf{X}, \mathbf{y} \sim \mathcal{N}(\sigma^{-2}\mathbf{K}^{-1}\mathbf{X}^T\mathbf{y}, \mathbf{K}^{-1})$$

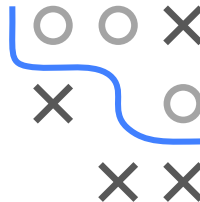
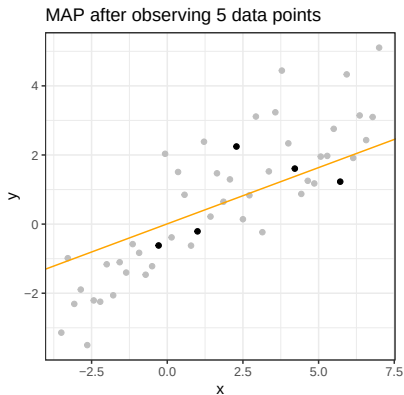
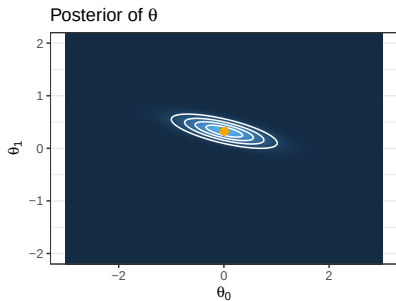
$$\text{with } \mathbf{K} := \sigma^{-2}\mathbf{X}^T\mathbf{X} + \tau^{-2}\mathbf{I}_p$$

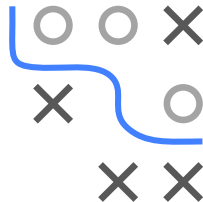
- Intuitively: quantifies posterior (i.e., after seeing data) probability of θ having generated the observed data



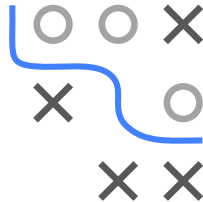
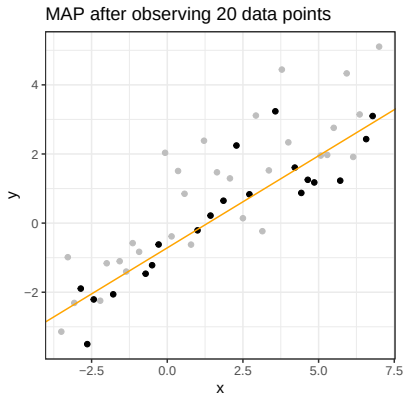
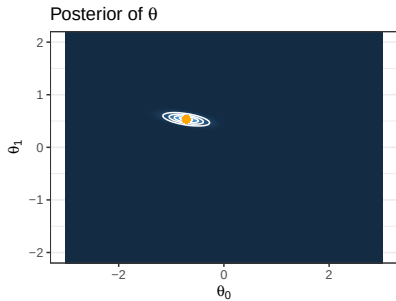


POSTERIOR CONTRACTION





POSTERIOR CONTRACTION



PROOF: GAUSSIANTY OF POSTERIOR

- We want to show that for

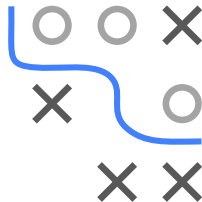
- Gaussian prior $\theta \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}_p)$ and
- Gaussian likelihood $\mathbf{y} \mid \mathbf{X}, \theta \sim \mathcal{N}(\mathbf{X}^T \theta, \sigma^2 \mathbf{I}_n)$

the resulting posterior is Gaussian: $\theta \mid \mathbf{X}, \mathbf{y} \sim \mathcal{N}(\sigma^{-2} \mathbf{K}^{-1} \mathbf{X}^T \mathbf{y}, \mathbf{K}^{-1})$

- Plug in Bayes' rule and keep only terms depending on θ

$$\begin{aligned} p(\theta \mid \mathbf{X}, \mathbf{y}) &\propto p(\mathbf{y} \mid \mathbf{X}, \theta) q(\theta) \propto \exp \left[-\frac{1}{2\sigma^2} (\mathbf{y} - \mathbf{X}\theta)^T (\mathbf{y} - \mathbf{X}\theta) - \frac{1}{2\tau^2} \theta^T \theta \right] \\ &= \exp \left[-\frac{1}{2} (\sigma^{-2} \mathbf{y}^T \mathbf{y} - 2\sigma^{-2} \mathbf{y}^T \mathbf{X}\theta + \sigma^{-2} \theta^T \mathbf{X}^T \mathbf{X}\theta + \tau^{-2} \theta^T \theta) \right] \\ &\propto \exp \left[-\frac{1}{2} (\sigma^{-2} \theta^T \mathbf{X}^T \mathbf{X}\theta + \tau^{-2} \theta^T \theta - 2\sigma^{-2} \mathbf{y}^T \mathbf{X}\theta) \right] \\ &= \exp \left[-\frac{1}{2} \theta^T \underbrace{(\sigma^{-2} \mathbf{X}^T \mathbf{X} + \tau^{-2} \mathbf{I}_p)}_{:=\mathbf{K}} \theta + \sigma^{-2} \mathbf{y}^T \mathbf{X}\theta \right] \end{aligned}$$

- Note how this resembles a normal density, except for term in orange
- (No need to worry about normalizing constant $\Rightarrow$ sole purpose: ensure density integrates to total prob of 1)



PROOF: GAUSSIANTITY OF POSTERIOR

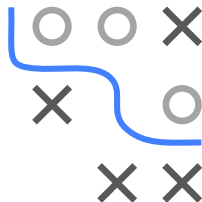
- Trick: introduce **constant** c , compensating for added quantities (“creative 0”), s.t. additions will conveniently cancel out with nuisance term

$$\begin{aligned} p(\theta | \mathbf{X}, \mathbf{y}) &\propto \exp\left[-\frac{1}{2}(\theta - c)^T \mathbf{K}(\theta - c) - c^T \mathbf{K} \theta + \underbrace{\frac{1}{2} c^T \mathbf{K} c}_{\text{doesn't depend on } \theta} + \sigma^{-2} \mathbf{y}^T \mathbf{X} \theta\right] \\ &\propto \exp\left[-\frac{1}{2}(\theta - c)^T \mathbf{K}(\theta - c) - c^T \mathbf{K} \theta + \sigma^{-2} \mathbf{y}^T \mathbf{X} \theta\right] \end{aligned}$$

- Choosing c s.t. $-c^T \mathbf{K} \theta + \sigma^{-2} \mathbf{y}^T \mathbf{X} \theta = 0$ leads to $\theta | \mathbf{X}, \mathbf{y} \sim \mathcal{N}(c, \mathbf{K}^{-1})$
- Using that $\mathbf{K}$ is symmetric, this implies

$$\begin{aligned} \sigma^{-2} \mathbf{y}^T \mathbf{X} &= c^T \mathbf{K} \\ \Leftrightarrow \sigma^{-2} \mathbf{y}^T \mathbf{X} \mathbf{K}^{-1} &= c^T \\ \Leftrightarrow c &= \sigma^{-2} \mathbf{K}^{-1} \mathbf{X}^T \mathbf{y} \end{aligned}$$

- Finally: $\theta | \mathbf{X}, \mathbf{y} \sim \mathcal{N}(\sigma^{-2} \mathbf{K}^{-1} \mathbf{X}^T \mathbf{y}, \mathbf{K}^{-1})$ $\square$



POSTERIOR PREDICTIVE DISTRIBUTION

- How does prediction change w.r.t. classical (non-Bayesian) LM?
- Gaussian posterior

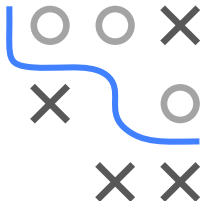
$$\boldsymbol{\theta} \mid \mathbf{X}, \mathbf{y} \sim \mathcal{N}(\sigma^{-2} \mathbf{K}^{-1} \mathbf{X}^T \mathbf{y}, \mathbf{K}^{-1})$$

induces (Gaussian) predictive distribution

- For a new observation $\mathbf{x}_*$ we get

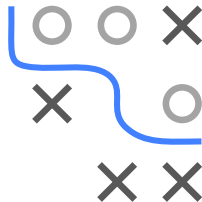
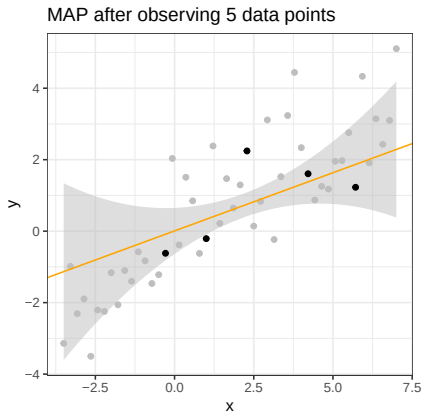
$$y_* \mid \mathbf{X}, \mathbf{y}, \mathbf{x}_* \sim \mathcal{N}(\sigma^{-2} \mathbf{y}^T \mathbf{X} \mathbf{K}^{-1} \mathbf{x}_*, \mathbf{x}_*^T \mathbf{K}^{-1} \mathbf{x}_*)$$

- Intuitively: expectation over all $\boldsymbol{\theta}$ -parameterized LMs, weighted according to posterior prob $\leftrightarrow$ classical LM: only max-prob $\boldsymbol{\theta}^{\text{MAP}}$
- Entire distribution with built-in uncertainty quantification!



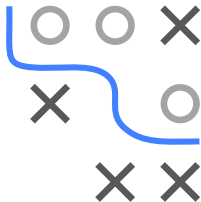
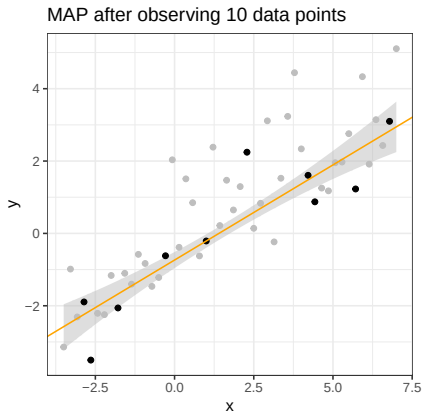
POSTERIOR MEAN AND VARIANCE

- For every test input $\mathbf{x}_*$, we get a posterior mean (orange) & variance (grey region; $\pm 2 \times$ standard deviation)



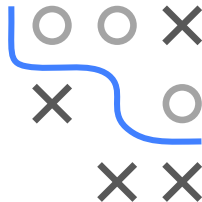
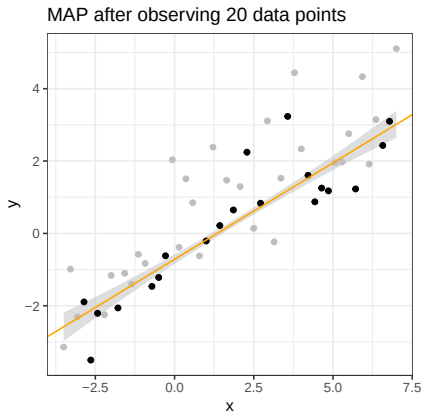
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POSTERIOR MEAN AND VARIANCE

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SUMMARY: BAYESIAN LM

- Bayesian perspective: entire distributions, rather than just point estimates, for θ
- From posterior distribution of θ we can derive a predictive distribution for $y_* = \theta^T \mathbf{x}_*$
- Online updates: after observing new data points, update posterior
 $\Rightarrow$ decreasing uncertainty

